Causal Benefit and Policy Risk Under Optional Assistance

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Abstract

An assistance offer can help people who accept it while harming people who would decline. Under one-sided uptake we derive the sharp range of outcomes from carrying out the intervention for everyone, and use it to bound the downside of an authorized policy change. For two voluntary offers we bound the benefit to newly induced takers when the offer itself can affect their outcomes. A separate result gives the possible shares of people helped and harmed, even when average benefit is positive. Exact finite checks reproduce the bounds. Applying them requires evidence for the direct-effect allowances and separate authorization for each regime. An exact allocation then balances worst-case policy regret against an overall individual-harm allowance across groups.

Summary and scope

Question. Does a favorable trial of voluntary help justify changing which people receive it?

Finding. With one-sided uptake, the unobserved outcomes of non-takers give a sharp interval for mandatory execution. With two voluntary offers, bounded direct effects give a different sharp interval for newly induced takers. Its two outcome marginals give a sharp joint region for the shares who benefit and who are harmed.

Evidence and scope. The bounds are proved and checked against exact synthetic oracles. An operator can report benefit sensitivity alongside uptake before expanding an offer. The required direct-effect bounds and individual consent come from outside the trial.

1 What a randomized offer identifies

A scheduling assistant can offer a change, let a person decline, and observe what happens. A positive result for the offer does not determine what would happen if the assistant carried out the change for everyone. The distinction is consequential when a service uses voluntary trial data to change an automated default.

Let Z be a randomized offer, $D(z)$ indicate uptake under offer status z, and $Y(d)$ be the outcome under treatment status d. Assume no interference, $D(1) \geq D(0)$, and that the offer affects the outcome only through uptake. Always-takers and never-takers contribute zero to the intent-to-treat contrast because their treatment does not change. Compliers contribute $Y(1) - Y(0)$. Thus, for nonzero uptake contrast,

$$\frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0]} = \mathbb{E}[Y(1) - Y(0) \mid D(1) > D(0)].$$

This is the local average treatment effect of Imbens and Angrist [2], in the instrumental-variable framework reviewed by Pearl [1]. The distinction matters for learned assistance policies because data from voluntary users may favor an action that other users reject.

A reversal in a finite population. In a finite population of 100 simulated recipients, 30 compliers gain two units and 70 never-takers lose two if treated. The uptake contrast is 0.30 and the offer's outcome contrast is 0.60, giving the ratio 2. Forcing treatment on everyone produces total change $60 - 140 = -80$. Both computations use exact integer or rational arithmetic.

2 What a favorable offer leaves unresolved

Consider one-sided uptake, so $D(0) = 0$ and $D(1) = 1$ for compliers only. Let c be the complier share. Outcomes lie in $[0, 1]$. Write $\mu_{d,C}$ and $\mu_{d,N}$ for mean outcomes under treatment d in compliers and never-takers. The offer arm identifies c, $\mu_{1,C}$, and $\mu_{0,N}$ when the corresponding strata have positive mass. The control arm identifies the population untreated mean W. This in turn identifies $\mu_{0,C}$ for $c > 0$. The policy that offers assistance has mean

$$O = c\mu_{1,C} + (1 - c)\mu_{0,N}.$$

Mandatory execution would yield $F = c\mu_{1,C} + (1 - c)\mu_{1,N}$, where $\mu_{1,N}$ is unobserved. Its sharp interval is therefore

$$[L, U] = [c\mu_{1,C},\ c\mu_{1,C} + 1 - c].$$

The endpoint worlds set the missing mean to zero or one without altering the observed arm distributions. Intermediate means are possible by mixing those worlds. The limiting case $c = 0$ uses $c\mu_{1,C} = 0$ and the case $c = 1$ collapses the interval. This is the one-sided special case of the partial-identification problem studied by Balke and Pearl [3].

3 A risk limit on policy experimentation

Choose the better of the two identified regimes and denote its mean by $S = \max(W, O)$. Consider a policy that uses the mandatory regime with probability θ and the identified regime otherwise, where $0 \le \theta \le 1$. This is an analysis of authorized regime choices. A standing refusal or missing authority excludes the mandatory regime before numerical optimization.

Suppose $L < S < U$. Define $a = S - L$ and $b = U - S$. The comparator knows F and chooses whichever regime has the higher mean. For $F \le S$, regret is $\theta(S - F)$ and increases as F decreases. For $F \ge S$, regret is $(1 - \theta)(F - S)$ and increases as F increases. The worst-case regret is therefore

$$R(\theta) = \max\{\theta a, (1 - \theta)b\}.$$

A separate constraint limits the worst-case expected loss against S to $\delta \ge 0$, which gives $\theta a \le \delta$. The unconstrained minimizer is the intersection of the two linear pieces. The objective decreases up to that point and increases after it. The constrained optimizer is

$$\theta^* = \min\left\{\frac{b}{a + b}, \frac{\delta}{a}\right\}.$$

If $L \ge S$ choose the mandatory regime whenever authorized. If $U \le S$ choose the identified regime. The formula applies the minimax-regret criterion for treatment decisions [4] with an additional downside constraint. It controls an expected policy contrast over the modeled population. In particular it cannot by itself grant individual permission to act.

4 Exact evaluation of the decision rule

Take $c = 2/5$, $\mu_{0,C} = 1/5$, $\mu_{1,C} = 4/5$, and $\mu_{0,N} = 3/5$. Offering improves the mean from $11/25$ to $17/25$. Mandatory execution can produce any mean in $[8/25, 23/25]$. Without the additional risk limit, $\theta^* = 2/5$. With $\delta = 1/20$, the feasible optimizer is $5/36$, with worst regret $31/150$. If mandatory execution is excluded by consent, the authorized policy set uses $\theta = 0$ and worst regret against the unrestricted oracle is $6/25$.

The checker evaluated 625 parameter settings and five loss caps per setting. The closed form attained no greater regret than any admissible member of a 17-point grid in all 3,125 cases. The risk cap reduced the unconstrained choice in 271 cases. Ninety parameter settings had strictly favorable offering and an interval for mandatory execution that straddled the offer's mean. These are finite design counts. The mathematical derivation covers continuous mixtures and the experiment checks its implementation.

5 Two voluntary offers with direct effects

An offer can change distress even when the recipient does not accept it. Instead of assuming exact exclusion, compare a weak and a strong authorized offer, labeled $z = 0, 1$. Keep randomized assignment, noninterference, and monotone uptake. The three uptake strata are stable takers A, induced takers C, and stable non-takers N, with shares a, c, n. They are identified from the two uptake rates, and assume $c > 0$. Write $Y_z(d)$ for a binary outcome under offer z and treatment d.

Use population-weighted success masses. Let A_z be the success mass for A under $Y_z(1)$ and N_z the mass for N under $Y_z(0)$. Let C_{zd} be the success mass for C under $Y_z(d)$. The weak treated cell identifies A_0, and the strong untreated cell identifies N_1. The other two cells identify $h_{11} = A_1 + C_{11}$ and $h_{00} = N_0 + C_{00}$. For example $h_{11} = \Pr(Y = 1, D = 1 \mid Z = 1)$.

Supply three sensitivity parameters, in population-mass units:

$$|A_1 - A_0| \le \epsilon_A, \qquad |N_1 - N_0| \le \epsilon_N, \qquad |C_{10} - C_{00}| \le \epsilon_C.$$

These are assumptions about direct offer effects at the same treatment status. The trial alone does not identify them. Define

$$\begin{aligned} I_A &= [A_0 - \epsilon_A, A_0 + \epsilon_A] \cap [0, a], \\ I_N &= [N_1 - \epsilon_N, N_1 + \epsilon_N] \cap [0, n], \\ I_1 &= (h_{11} - I_A) \cap [0, c], \\ I_0 &= (h_{00} - I_N) \cap [0, c], \\ J_0 &= (I_0 + [-\epsilon_C, \epsilon_C]) \cap [0, c]. \end{aligned}$$

Here sums and differences range over all points in the intervals.

Theorem 1. *If either I_1 or I_0 is empty, the observed cells are incompatible with the stated bounds. Otherwise the sharp identified interval for the treatment benefit among induced takers under the strong offer is*

$$\mathbb{E}[Y_1(1) - Y_1(0) \mid C] \ \in \ \frac{1}{c}(I_1 - J_0).$$

Proof. The strong treated mixture gives $C_{11} = h_{11} - A_1$, and the weak untreated mixture gives $C_{00} = h_{00} - N_0$. This yields I_1 and I_0 by the observed totals and the direct-effect bounds. The constraint on the untreated induced stratum expands I_0 to J_0. The choices of C_{11} and C_{10} are

independent as feasibility constraints: they involve disjoint unknown masses and can be combined in one joint potential-outcome law with the specified marginals. Thus every pair of points in I_1 and J_0 is feasible, including the pairs that attain the endpoints. □

For a trial with $(a, c, n) = (1/4, 1/4, 1/2)$, suppose $(A_0, h_{11}, h_{00}, N_1) = (0.15, 0.35, 0.35, 0.25)$. The uptake contrast is $1/4$ and the outcome contrast is 0.10. Under exact exclusion their ratio gives benefit 0.40. Allowing same-treatment mean shifts of 0.10 for A and N and 0.05 for untreated C gives benefit in $[0.05, 0.75]$. Increasing only the allowable mean shift for N to 0.20 widens the interval to $[-0.15, 0.90]$.

An exact integer-grid audit enumerated latent success counts and independently compared both endpoints or infeasibility for 56,700 observed count and sensitivity settings. It found 4,299 settings with a positive conventional Wald contrast and a strictly sign-ambiguous interval. That count describes the declared grid. The sharpness argument works over continuous population masses and does not rely on the grid.

6 A positive mean can conceal individual harm

For the strong offer and induced takers, write $[p_-, p_+] = I_1/c$ for the treated success probability and $[q_-, q_+] = J_0/c$ for the untreated one. A person benefits when the treated outcome is one and the untreated outcome is zero, and is harmed in the reverse case. Let b and h be the corresponding population shares within the induced stratum. The mean treatment effect is $b - h$. It leaves the two shares underdetermined.

Theorem 2. *With unrestricted dependence of the two potential outcomes, the sharp region for* (b, h) *is*

$$\begin{aligned} 0 \le b &\le \min(p_+, 1 - q_-), \\ 0 \le h &\le \min(q_+, 1 - p_-), \\ b + h &\le 1, \\ p_- - q_+ \le b - h &\le p_+ - q_-. \end{aligned}$$

The sharp marginal lower bounds are $\max(0, p_- - q_+)$ *for* b *and* $\max(0, q_- - p_+)$ *for* h*. The first two inequalities give their respective sharp upper bounds.*

Proof. Let u be the share with both potential outcomes equal to one. The four joint masses are $b, h, u, 1 - b - h - u$. With $b, h \ge 0$, their nonnegativity and the marginal constraints hold exactly when

$$[0, 1 - b - h] \cap [p_- - b, p_+ - b] \cap [q_- - h, q_+ - h]$$

is nonempty. Requiring each lower endpoint to be at most each upper endpoint gives the displayed inequalities. Any allowed u constructs a joint law, which can be coupled with the remaining binary potential outcomes to match their feasible marginals in the preceding theorem.

For fixed success marginals p, q, the smallest and largest possible benefit shares are $\max(0, p-q)$ and $\min(p, 1-q)$. This follows by choosing the largest and smallest feasible common-success share u, respectively. Minimizing the former over the marginal intervals uses p_-, q_+. Maximizing the latter uses p_+, q_-. Interchanging the roles of treatment and no treatment proves the harm bounds. □

In the trial with the narrower direct-effect allowances, the induced group's mean treatment effect lies in $[0.05, 0.75]$, its benefited share in $[0.05, 0.85]$ and its harmed share in $[0, 0.30]$. A joint

law with $b = 0.35$, $h = 0.30$ and $u = 0.35$ matches the observed cells and attains the lowest mean effect. The largest benefit and harm bounds cannot occur together because their sum would exceed one. Thus an operator with a ten-percent ceiling on individual harm cannot certify that ceiling from the positive mean alone.

The verifier enumerated joint outcome tables on populations of one to eight induced takers. The region and all four coordinate extrema matched direct enumeration for 4,916 pairs of marginal intervals and 308,846 candidate benefit and harm pairs. Of those interval pairs, 560 permit positive harm even though every compatible law has positive mean benefit. This is a finite algebra check. Trial sampling uncertainty and the direct-effect assumptions still need separate analysis in an application.

7 Allocating authorized help under a harm allowance

Suppose a service has pre-decision groups with shares w_g that sum to one. For each group, assume justified treatment-effect bounds $[\ell_g, u_g]$ and a sharp upper bound $\bar{h}_g$ on the share of people whose outcome would worsen under treatment. For binary outcomes, the preceding joint region supplies these quantities. Take the joint uncertainty across groups to be the product of their admissible laws. Compare treatment with a fixed reference under the same offer. Within each group the action is authorized, and the service assigns treatment with probability θ_g independently of each person's potential outcomes.

Let $a_g = [-\ell_g]_+$ and $b_g = [u_g]_+$, where $[x]_+ = \max(x, 0)$. An oracle that knows the mean effect can choose to treat a whole group or leave it at the reference. The worst regret against this oracle and the worst share of people harmed are, respectively,

$$\sum_g w_g \max\{a_g\theta_g, b_g(1-\theta_g)\}, \qquad \sum_g w_g \bar{h}_g \theta_g.$$

For the first expression, consider the extreme group means. For the second, independence of assignment gives a harm share of $\theta_g h_g$ in each group. Product uncertainty makes both bounds sharp. The two suprema can occur in different worlds.

Theorem 3. *To minimize worst-case regret subject to an overall harm allowance η, let $x_g = b_g/(a_g + b_g)$ when the denominator is positive, and set $x_g = 0$ otherwise. Use $\theta_g = x_g$ for groups with $\bar{h}_g = 0$. For the others, assign treatment probability up to x_g in descending order of $b_g/\bar{h}_g$, stopping when $\sum_g w_g \bar{h}_g \theta_g = \eta$ or all group limits are reached.*

Proof. Moving any $\theta_g > x_g$ down to x_g weakly lowers both the regret and its harm cost. On $[0, x_g]$, increasing θ_g saves $w_g b_g$ units of regret per unit of probability and costs $w_g \bar{h}_g$ units of the harm allowance. There is no cost when $\bar{h}_g = 0$. For positive cost, moving an equal amount of harm allowance from a lower-ratio group to a higher-ratio group with room left cannot increase regret. Repeating the exchange gives the stated allocation, including ties and a partial final group. □

For two equally sized groups, take treated and untreated success intervals $[0.40, 0.80]$ and $[0.20, 0.40]$ in the first group, and $[0.90, 0.90]$ and $[0.60, 0.80]$ in the second. The upper mean improvements are 0.60 and 0.30. The maximum harmed shares are 0.40 and 0.10. At overall allowance 0.08, serving the second group first gives treatment rates $(0.15, 1)$ and regret 0.255. Serving the larger upper mean first gives rates $(0.40, 0)$ and regret 0.330. Both meet the same harm allowance.

The verifier reconstructed 384 group extrema from joint outcome tables and matched 640 allocations against a vertex oracle for the piecewise-linear objective. In an application, use group

bounds for the actual authorized population. Principal strata from a trial can support a sensitivity analysis, but observed group membership needs its own evidence. The overall allowance can be supplemented by separate harm ceilings in groups requiring additional protection.

8 Application and next test

Train the policy to distinguish offering, accepting, and executing. During a controlled evaluation of an accessibility assistant, log opt-out friction and downstream outcomes separately. If receiving an offer itself changes distress, exclusion fails and the randomized effect of the offer remains the defensible estimand. Cross-user coordination can also violate noninterference. A follow-up should test a low-friction offer against a matched control and report both uptake and outcome contrasts. A negative offer effect would directly challenge the proposed deployment even with favorable outcomes among acceptors.

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